

Entra Join, Autopilot & Intune - Cloud-Native Windows Endpoint Playbook

Purpose. A complete procedure for provisioning a business Windows laptop as a cloud-native endpoint: Microsoft Entra joined, Intune managed, encrypted, with a managed local administrator account and a standard-user daily account. The device is built through Windows Autopilot device preparation, with device association used to mark the device as corporate-owned without registering a hardware hash.

Reader. A technician who has administrator access to a Microsoft 365 tenant and a laptop to provision, and who has not done this before.

Reference build. Every step was executed on a Lenovo ThinkPad E16 Gen 2 (AMD, machine type 21M5) bought from a third-party marketplace reseller, in a Microsoft 365 Business Premium tenant. Where the device or the product behaved differently from the documentation, the step says so in a **Reference build** note. Those notes are the reason this playbook exists.

Placeholders. `<ORG>` is a short organization prefix. `contoso.com` is the tenant domain. `user@contoso.com` is the person who will use the device. Replace all three before use.

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Written against Windows 11 Pro, Windows Autopilot device preparation with device association (introduced 27 August 2026), Microsoft 365 Business Premium

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Companion case study evanireland.tech/projects/entra-autopilot-intune-endpoint/

Scripts evanireland.tech/reference/scripts/

Contents

1. Concepts & Terminology
2. Prerequisites, Kit & Decisions
3. Tenant Preparation
4. Device Intake & Firmware
5. Clean Install
6. Device Association & Provisioning
7. Verification
8. User Applications Through Company Portal
9. Hardening Backlog
10. Troubleshooting Reference
11. Verification Checklist

Sources follow Part 11.

1. Concepts & Terminology

1.1 Join Types

The phrase "Entra domain join" does not describe a real configuration. The correct term is **Microsoft Entra join**. There is no domain, no domain controller, no Group Policy, and no network line of sight to anything on premises. The device trusts the tenant and receives a device certificate and a Primary Refresh Token (PRT) from Entra ID.

Join type	Identity authority	Management	Use when
Domain join	On-premises Active Directory	Group Policy, Configuration Manager	An on-premises domain exists and the device must authenticate to it
Microsoft Entra hybrid join	On-premises AD and Entra ID together	Group Policy and Intune	A domain exists and cloud management is being added
Microsoft Entra join	Entra ID only	Intune	No on-premises domain exists. This playbook

1.2 Windows Autopilot Device Preparation

The Microsoft name is **Windows Autopilot device preparation** (APDP), informally "Autopilot v2". Classic Windows Autopilot (v1) is not deprecated. It still owns pre-provisioning, self-deploying mode, and hybrid join, none of which APDP supports.

Classic Autopilot depends on the device's **hardware hash** being registered in the tenant, normally by the OEM or a reseller at the point of sale. A laptop bought from a third-party marketplace seller has no such registration. APDP needs none.

Device association closes most of the gap that hash registration used to fill. The device exports identifying information, an administrator uploads it, and the device writes a TPM-attested tenant marker into UEFI. The result is corporate ownership, a device name template, device-targeted policy assignment, and a shortened out-of-box experience (OOBE).

Requirements, from Microsoft Learn: Windows 11 24H2 or 25H2 with update KB5120998 or later, a physical device, TPM 2.0, and a UEFI that supports the marker. Virtual machines do not qualify.

Reference build. Association worked on the first attempt and the name template applied. It worked only over Ethernet. On Wi-Fi the "Assign device association" option forced a wired connection before it would continue. Treat a wired connection as mandatory.

1.3 Enrollment Time Grouping

APDP places the new device into a security group during enrollment, and everything assigned to that group begins to apply at once. A dynamic group would have to wait for rule evaluation, which can take many minutes.

Intune can write that group membership only when the **Intune Provisioning Client** service principal (App ID `f1346770-5b25-470b-88bd-d5744ab7952c`) is an **owner** of the group. Owner rights on one group is a narrower grant than tenant-wide group write permission.

1.4 Provisioned and Installed Packages

A **provisioned** package is the template Windows copies into every new user profile at first sign-in. An **installed** package is a copy already registered to an existing profile. Removing only the installed copy lets the app return for the next new profile. Removing both, during OOBE and before the first profile exists, means the app never appears.

2. Prerequisites, Kit & Decisions

2.1 Licensing

Microsoft 365 Business Premium carries three pieces this build needs.

Component	Used for
Intune Plan 1	Mobile device management enrollment and policy
Microsoft Entra ID P1	Automatic MDM enrollment, local administrator password recovery
Windows 11 Business rights	Subscription activation

APDP fails silently when the signing-in user holds no Intune license.

Subscription activation is a step-up, not a base license. Business Premium upgrades a device that is already licensed for Windows 11 Pro. It does not license a device that has no valid Windows license underneath. Part 4 covers how to establish that before the wipe.

Windows edition. Business Premium licenses Windows 11 **Pro**. Two policies that would otherwise simplify this build apply only to Enterprise and Education: the in-box app removal policy, and "Turn off Microsoft consumer experiences". On Pro they report "Not applicable". Section 3.6 replaces them with a script.

2.2 Kit

Item	Purpose
Administrator workstation	Portal work, USB media creation
USB stick A, 8 GB or larger	Windows installer. The stick is erased
USB stick B, NTFS	Scripts in, inspection reports and the device association export out. Microsoft recommends NTFS for the export
Ethernet cable, or a USB-C Ethernet adapter	Wired connection for OOB. Wired is required for device association in practice
Phone	Photographs of screens that cannot be captured, such as BIOS and OOB
Authenticator	Multifactor sign-in for the build user

2.3 Decisions

Change these before starting, not partway through.

Decision	Choice	Reason
Keep the seller image or reinstall	Clean install from Microsoft media	An unknown party built the image. A Windows Reset restores the seller's customizations from the recovery partition. A clean install is also the debloat
Enrollment method	APDP, user-driven, with device association	No hash registration is available. Association adds naming and corporate marking
Fallback if association fails	APDP by user group assignment, rename afterward	Same result without the name template
Join type	Microsoft Entra join	APDP supports no other type, and no domain exists
Local account type in policy	Standard user	Correct template for client devices
Device name	<ORG>-%SERIAL%	Ties the name to the asset record. Keep the result at or under 15 characters, the NetBIOS limit
Group naming	SG-APDP-Users , SG-APDP-Devices	Both assigned . APDP requires the device group to be assigned, not dynamic
Apps during setup	Company Portal, Microsoft 365 Apps	Only what must exist before first use. Every added app is a way for setup to fail
Office update channel	Current Channel	Copilot features ship there first
BitLocker	XTS-AES 256, silent, key escrowed to Entra	Verify after setup. See Section 7.5
Local administrator	Windows LAPS, automatically managed custom account	Built-in Administrator is disabled by default on Windows 11
Windows license	Verified before the wipe, retail Pro key if none is valid	See Section 4.5
User applications	Available in Company Portal after the desktop	Nothing optional is allowed to fail OOBE

2.4 Time Budget

Part	Work	Estimate	Safe stopping point
3	Tenant preparation	60 to 90 minutes	Yes. Nothing touches the laptop
4	Intake, BIOS, inspection, decision gate	30 to 45 minutes	Yes. The laptop is still in the seller's state
5	Clean install	20 to 30 minutes	Only at the OOBE region screen
6	Association, OOBE, APDP	30 to 60 minutes	No. Run through to the desktop
7	Verification	45 to 60 minutes	Yes, between checks
8	User applications	Separate session	Yes

Reference build. About 9.5 hours over two sessions. Tenant preparation took about 2 hours including two policy correction passes. Intake grew by about 2 hours when the decision gate failed and a firmware scrub and secure wipe were added. APDP ran from association to desktop in 14 minutes, of which the provisioning run itself was about 9. Verification took about 4 hours, mostly the BitLocker cipher fix, a license purchase, a BIOS update, and application deployment.

3. Tenant Preparation

All of Part 3 happens in a browser on the administrator workstation. Nothing touches the laptop.

Portal paths change frequently. The paths below were current in September 2026. If one has moved, search the admin center for the feature name.

Sign in as a Global Administrator or equivalent to the **Entra admin center** (entra.microsoft.com) and the **Intune admin center** (intune.microsoft.com).

3.1 License Confirmation

Where: Microsoft 365 admin center > Users > Active users > the build user > Licenses and apps.

Confirm: Microsoft 365 Business Premium is assigned. Add Copilot if the tenant licenses it.

3.2 Entra Device Settings

Where: Entra admin center > Entra ID > Devices > All devices > Device settings.

Setting	Value	Reason
Users may join devices to Microsoft Entra	All, or Selected including the build user	The first user to sign in performs the join. If the user is not allowed, OOBЕ fails with error 801C03ED
Require multifactor authentication to register or join devices	No	Microsoft guidance is to enforce MFA for join through Conditional Access instead of this switch
Maximum number of devices per user	Default (50)	No change needed
Registering user is added as local administrator during join	None	Works with the APDP standard-user setting. Does not affect the Global Administrator role
Global administrator role is added as local administrator during Microsoft Entra join (preview)	No	This is the setting that controls the Global Administrator role. With it on, every Global Administrator is a local administrator on every joined device
Enable Microsoft Entra Local Administrator Password Solution (LAPS)	Yes	Required for the Intune LAPS policy in Section 3.8 to back up passwords. Without it, LAPS fails without any error

Save and capture a screenshot.

Reference build. The Global Administrator toggle was not in the original plan, which assumed the role was always a local administrator. It is not. With the toggle set to No and the APDP standard-user setting, the signed-in Global Administrator was a standard user on the laptop, and every elevation went through the LAPS-managed account.

3.3 Automatic MDM Enrollment Scope

Where: Intune admin center > Devices > Enrollment > Windows > Automatic Enrollment. The same page is reachable from Entra ID > Mobility (MDM and WIP) > Microsoft Intune.

Setting	Value
MDM user scope	All, or Some with SG-APDP-Users once Section 3.4 exists
MAM user scope	None

This is the link that turns "Entra joined" into "Intune managed". When a user in MDM scope joins a device to Entra, Windows enrolls it into Intune using the MDM URLs on this page. The `enterpriseenrollment` and `enterpriseregistration` DNS CNAME records do not take part in this path. They serve manual enrollment by email address.

3.4 Groups

Where: Entra admin center > Groups > All groups > New group.

Users group. Type Security. Name `SG-APDP-Users`. Membership type **Assigned**. Members: the build user. The APDP policy targets this group.

Devices group. Type Security. Name `SG-APDP-Devices`. Membership type **Assigned**. Members: none.

Owners: add Intune Provisioning Client. Search by name or by App ID `f1346770-5b25-470b-88bd-d5744ab7952c`. In some tenants the same service principal displays as **Intune Autopilot**

ConfidentialClient. Same App ID, same object.

Do not select **Intune CMDeviceService**. It is a different application.

The owner step is what makes enrollment time grouping work (Section 1.3).

If the owner search finds nothing, the service principal does not exist in the tenant yet. Create it from PowerShell on the administrator workstation:

```
Install-Module Microsoft.Graph.Applications -Scope CurrentUser
Connect-MgGraph -Scopes "Application.ReadWrite.All"
New-MgServicePrincipal -AppId "f1346770-5b25-470b-88bd-d5744ab7952c"
```

- `Install-Module ... -Scope CurrentUser` downloads the Microsoft Graph module into the current user's profile only. No administrator rights are needed on the workstation.
- `Connect-MgGraph -Scopes ...` signs in to Microsoft Graph and requests exactly one permission: create and modify application objects. Consent is given in the browser.
- `New-MgServicePrincipal -AppId ...` creates the tenant-local instance of Microsoft's multi-tenant Intune provisioning application. The application exists globally. The tenant was missing only its local representation.

Retry the owner search afterward.

Reference build. In a new tenant the service principal did not exist. Neither the name nor the App ID returned a result. The three commands fixed it. Treat this as standard preparation for every new client tenant.

3.5 Applications for Setup

Where: Intune admin center > Apps > Windows > Create.

Company Portal. App type Microsoft Store app (new). Search `Company Portal`. Install behavior **System**. Assignment **Required** to `SG-APDP-Devices`. Company Portal is the self-service front end for Intune. APDP supports only the "new" (WinGet-based) type of Store app.

Microsoft 365 Apps. App type Microsoft 365 Apps, Windows 10 and later. Configuration designer. Apps: Excel, OneNote, Outlook, PowerPoint, Teams, Word. Leave Access, Publisher and Skype for Business off. Architecture 64-bit. Update channel Current Channel. Remove other versions Yes. Accept the license agreement on behalf of users Yes. Shared computer activation No. Default file format Office Open XML. Assignment **Required** to `SG-APDP-Devices`.

The apps are required for the device group, not the user group, because APDP installs them during OOB before any user is signed in, in system context.

The APDP limit is 25 applications. Keep the real number between two and five. Install everything else after the desktop (Part 8).

3.6 Debloat Script

A clean install from Microsoft media removes all OEM and reseller software. What remains is Microsoft's own consumer apps: Solitaire, the Xbox family, News, Clipchamp and similar. A short, exact-name, logged removal script handles them.

Do not run a general-purpose debloat tool on a managed endpoint. Tools of that type change services, telemetry, Edge, and Copilot toggles, which conflicts with Intune and Defender, can break features that depend on diagnostic data, and cannot be repeated or audited across clients. The skill is debloat as code through the management platform.

Script. `Remove-Bloat.ps1`, published at evanireland.tech/reference/scripts/. Read it before use. The only part to edit is `$RemoveList` near the top, and `$OrgName`, which names the log folder under `C:\ProgramData`.

Behavior.

- Allow-list by exact package name. No wildcards, because a pattern such as `*Xbox*` can match packages that were not meant to be removed.
- A protected list that the script refuses to remove even if a name is added to the removal list by mistake: Copilot, the Microsoft 365 Copilot app, Teams, Outlook, To Do, Power Automate, Phone Link, Quick Assist, winget (App Installer), the Store, Windows Security, Company Portal.
- Removes both the provisioned and the installed copy of each package (Section 1.4).
- Idempotent. A package that is already gone is logged and skipped.
- Writes a full transcript to `C:\ProgramData\<OrgName>\Logs\` as proof.

- Always exits 0. A leftover game must not make Intune report the whole provisioning as failed. Failures are reviewed in the log.

Upload. Intune admin center > Devices > Scripts and remediations > Platform scripts > Add > Windows 10 and later.

Setting	Value	Reason
Name	<ORG> - Remove consumer inbox apps	
Run this script using the logged on credentials	No	Runs as SYSTEM. Required during OOBE, when no user is signed in
Enforce script signature check	No	
Run script in 64-bit PowerShell host	Yes	The Appx cmdlets misbehave in the 32-bit host
Assignment	SG-APDP-Devices	

Limits. A platform script runs once per device and runs again only when a changed script is uploaded. Windows feature updates can re-add inbox apps. Upload a new version of the script after a feature update if apps return.

Reference build. The script worked as written. One consumer app, Weather, was removed from the list by choice. A news feed still appeared in the Widgets board after the News app was removed. The feed belongs to Widgets, not to the app. Hide Widgets on the taskbar or turn it off by policy (Section 9). Removing Solitaire and Xbox from a machine with a Ryzen 7 and 32 GB of RAM produces no measurable performance change. The gain is a clean, known image and the absence of OEM services.

3.7 Disk Encryption Policy

Where: Intune admin center > Endpoint security > Disk encryption > Create policy > Windows > BitLocker. Name `<ORG> - BitLocker - OS drive`.

Setting	Value	Reason
Require Device Encryption	Enabled	Turns BitLocker on
Allow Warning For Other Disk Encryption	Disabled	Required for silent encryption without a user prompt
Allow Standard User Encryption	Enabled	Lets encryption proceed without an administrator session
Configure Recovery Password Rotation	Refresh on for Azure AD-joined devices	A used recovery key is replaced automatically
Choose drive encryption method and cipher strength	OS drive XTS-AES 256, fixed drive XTS-AES 256, removable drive AES-CBC 128	256-bit on internal drives. AES-CBC on removable media so it stays readable on older machines
Require additional authentication at startup	Enabled. TPM: Require TPM. Startup PIN, key, and key plus PIN: Do not allow	TPM-only unlock makes silent encryption possible. A pre-boot PIN is stronger and is a later decision
Choose how BitLocker-protected OS drives can be recovered	Enabled. Allow 48-digit recovery password. Save BitLocker recovery information to AD DS: True. Do not enable BitLocker until recovery information is stored: True	"AD DS" means Entra ID on an Entra-joined device. The last setting is the important one: no escrowed key, no encryption

Assign to `SG-APDP-Devices`. Before saving, confirm that "Do not enable BitLocker until recovery information is stored" reads **True**.

Reference build. This policy reported **Succeeded** on every setting while the drive sat at XTS-AES 128. Windows automatic device encryption started before the policy arrived, and a cipher setting never re-encrypts a drive that is already encrypted. "Succeeded" means the setting reached the registry. It does not mean the disk matches it. Microsoft documented that, from 14 September 2026, APDP no longer forces its default 128-bit policy during OOB. The reference drive still encrypted at 128 bits first. Section 7.5 has the check and the fix. In the same build, the recovery-information setting was saved as False on the first pass.

3.8 Local Administrator Password Policy

Where: Intune admin center > Endpoint security > Account protection > Create policy > Windows > Local admin password solution (Windows LAPS). Name <ORG> - LAPS .

Setting	Value	Reason
Backup Directory	Backup the password to Azure AD only	Entra-joined device, no on-premises AD
Password Age Days	30	Rotation interval. The Entra minimum is 7
Automatic Account Management Enabled	Enabled	Windows 11 24H2 and later. The built-in Administrator is disabled by default, so pointing LAPS at it rotates the password of an account nobody can use. This setting makes LAPS create and own a dedicated account
Automatic Account Management Target	Manage a new custom administrator account	
Automatic Account Management Name Or Prefix	<org>-lapsadmin	Avoid admin and administrator . They are the first names an attacker tries
Automatic Account Management Enable Account	Enabled	
Automatic Account Management Randomize Name	Disabled	Predictable name while learning the process
Password Complexity	Passphrase (long words)	The password is typed by hand at a sign-in screen. Six long words give strength with a lower typing error rate than 20 random symbols
Passphrase Length	6	Leave Password Length not configured when a passphrase is used
Post Authentication Actions	Reset password	The password is replaced after use without ending an administrator session mid-task
Post Authentication Reset Delay	24 hours	One retrieved password covers a working day

If a setting name differs in the portal, the four Automatic Account Management settings are the ones that matter. Assign to SG-APDP-Devices .

Reference build. The first draft of this policy was saved without Automatic Account Management. Check those four rows before saving.

Why these two policies now. Policies assigned to the device group synchronize during setup. The device reaches the desktop already encrypted and with a managed break-glass local administrator, instead of an unprotected window that is closed later.

3.9 Company Branding (Optional)

Where: Entra admin center > User experiences > Company branding > Default sign-in > Edit.

Add a square logo (240 by 240), a banner logo (280 by 60), and sign-in page text. The branding appears during OOB sign-in. It is also a prerequisite for the "Hide change account options" setting in Section 3.10. It takes about 15 minutes and is a visible result on every client engagement.

3.10 Device Preparation Policy

Where: Intune admin center > Devices > Windows > Enrollment > Windows Autopilot device preparation > Device preparation policies > Create > User Driven.

Setting	Value
Name	<ORG>-APDP-UserDriven-v1
Device group	SG-APDP-Devices (the device group, not the user group)
Deployment mode / type	User-driven / Single user
Join type	Microsoft Entra joined
User account type	Standard User
Minutes allowed before showing installation error	60
Custom error message	Setup did not finish. Contact <ORG> support: support@contoso.com
Allow users to skip setup after multiple attempts	No
Show link to diagnostics	Yes
Apps	Company Portal, Microsoft 365 Apps
Scripts	<ORG> - Remove consumer inbox apps
Assignments	SG-APDP-Users (the user group)

"Allow users to skip" is set to No because a failure should be seen and fixed, not skipped into a half-configured desktop.

Settings that apply only on associated devices (Part 6):

Setting	Value
Language (Region)	Your region
Automatically configure keyboard	Yes
Hide Microsoft Software License Terms	Yes
Hide privacy settings	Yes. Location services default to off, so set the time zone by hand in Section 7.9
Hide change account options	Yes only if Section 3.9 is done. Otherwise No
Apply device name template	Yes, <ORG>-%SERIAL%

The chain this tenant now implements. This is the answer to "walk through Autopilot device preparation":

1. The user signs in during OOB.

2. The device **Entra joins**: a device object, a certificate, and a PRT.
3. The user is in MDM scope, so Windows **auto-enrolls in Intune**.
4. Intune finds the APDP policy assigned to the user's group.
5. **Enrollment time grouping** places the device in the device group.
6. The Intune Management Extension installs the selected apps and runs the selected scripts while OOBЕ shows a percentage.
7. Everything else assigned to the device group, including BitLocker and LAPS, synchronizes.
8. The desktop appears.

Stopping point. Nothing so far has touched the laptop.

4. Device Intake & Firmware

4.1 Unbox & Document

Photograph the box label, the bottom-of-laptop sticker (machine type model and serial number), and anything the seller added: stickers, seals, paperwork. Record the model and serial number.

Note whether the bottom panel shows signs of having been opened. On a reseller-upgraded unit it will have been. The purpose is a record, not an accusation. The same habit applies to every client device received.

4.2 Warranty & Original Configuration

On the administrator workstation, enter the serial number at pcsupport.lenovo.com. Open Warranty and Product specifications. Record the warranty type and end date, and the original factory configuration, especially the operating system the unit shipped with. The shipped operating system is an independent confirmation of the firmware license (Section 4.5).

Reference build. Depot warranty, ending about six months out, with a ship-to location in another country. Not on-site, nearly expired, and registered abroad: the first sign the unit was a grey-market import.

4.3 BIOS & Firmware

Make every firmware change before device association. Association is invalidated if BIOS settings are later reset or the Secure Boot configuration changes. Make all firmware changes first, then leave the firmware alone.

Power on and press **F1** at the logo. Do not let the machine boot Windows yet. Photograph the BIOS as received before changing anything.

Where (Lenovo BIOS)	Check or set	Reason
Main	Record BIOS version, model, serial, installed memory	Confirms the seller's memory claim at the hardware level
Security > Secure Boot	On, not in Setup Mode	Windows 11 requirement. Association checks it
Security > Security Chip	Enabled. Note Discrete TPM 2.0 or Microsoft Pluton	Association attests identity with TPM 2.0
Security > Virtualization	AMD-V (SVM) On	Needed for virtualization-based security and for Hyper-V later
Startup	UEFI only, boot mode normal	
Date and time	Correct	Clock skew breaks token issuance
Security > Password > Supervisor password	Set it, after storing it in a password manager	Prevents booting a USB device or changing Secure Boot without it

Lenovo supervisor passwords are not recoverable. If the password is lost, the fix is a system board replacement. Store the password first. Set it second.

Scrub for an untrusted unit. When the device came from a seller whose image cannot be trusted, add these steps before the clean install:

Step	Where	Reason
Clear the TPM	Security > Security Chip > Clear Security Chip	Removes keys or ownership left by the previous builder. Association attests with a fresh TPM
Load defaults	F9, or Restart > Load Factory Defaults	Undoes settings the seller changed
Re-apply the table above		Defaults do not guarantee Secure Boot, SVM, UEFI-only, or the clock
Set the supervisor password	Security > Password	Required by the next step
Secure Wipe	Security > Secure Wipe (Data Wipe)	Prompts for the supervisor password. Use the single-pass full-erase method

Single pass is correct for a solid-state drive. Multi-pass overwriting such as DoD 5220.22-M was designed for magnetic disks. An SSD controller remaps writes across flash cells, so extra passes add wear without reaching more data. The drive's own erase is the NIST SP 800-88 "clear" equivalent.

Photograph each screen. Press **F10** to save and exit.

Reference build. The unit arrived with Secure Boot off. The firmware was scrubbed as above after the inspection and decision gate in Sections 4.4 and 4.5.

4.4 Pre-Flight Inspection

Read the image before wiping it. **Leave the Ethernet cable unplugged and do not join Wi-Fi.** The image is being read, not given a chance to do anything online.

- Let the machine boot. It lands in one of two places:
 - The OOBЕ region screen** (most likely). Press **Shift+F10**. If nothing happens, try **Fn+Shift+F10**, because ThinkPad function keys default to media keys. **Fn+Esc** toggles FnLock. A black Command Prompt opens with administrator rights, before any account exists.
 - A desktop**, because the seller finished OOBЕ. Right-click Start and open Terminal (Admin).
- Insert stick B. Find its drive letter:

```
powershell -NoProfile -Command "Get-Volume | Format-Table  
DriveLetter,FileSystemLabel,FileSystem,SizeRemaining"
```

- Run the inspection script, replacing **E:** with the drive letter:

```
powershell -NoProfile -ExecutionPolicy Bypass -File E:\Inspect-Laptop.ps1
```

- `powershell` starts Windows PowerShell 5.1 from the Command Prompt.
- `-NoProfile` skips profile scripts.
- `-ExecutionPolicy Bypass` applies to this one process only, so an unsigned script runs without changing the machine's policy.
- `-File E:\...` is the script to run. It only reads. It prints a report and saves a copy as `Inspection-<name>-<timestamp>.txt` on stick B.

4. Photograph the screen as a backup. Run `exit` and hold the power button to shut down. Do not complete OOBE on the seller image.

The script `Inspect-Laptop.ps1` is published at evanireland.tech/reference/scripts/. It changes nothing. It reports hardware identity, the Windows edition as installed, the firmware license versus the installed license, license channel and activator traces, seller customizations that a Windows Reset would not remove, memory and storage, TPM and Secure Boot state, network adapters, and devices with driver problems.

4.5 Decision Gate

Read Section 3 of the report (licensing) and decide.

Report says	Meaning	Action
Firmware key present, edition Professional	The board is licensed for Pro. A clean Pro install activates from firmware	Proceed to Part 5
Firmware key present, edition Core (Home)	The board is licensed for Home. The listing's "Pro" came from the seller's own key	Stop. Home cannot Entra join. Return or dispute with the seller, or buy a legitimate Pro upgrade
No firmware key, installed channel Retail or OEM , status Licensed	The seller added a license. It may reactivate after a clean install through a digital license tied to the hardware, or it may not	Proceed with activation risk. Ask the seller for license documentation while the return window is open
Installed channel Volume:GVLK or KMS client, a KMS host listed, or activator traces	Pirated activation	The image is compromised. Wipe it and treat the license as void. A legitimate Pro license is required before this is a business machine
No firmware key, installed channel VOLUME_MAK , status Licensed, no KMS host, no activator traces	A volume license issued to an organization, placed on a laptop sold to an individual. Activated is not the same as licensed	Keep the hardware, scrub, clean install, and buy a retail Windows 11 Pro key after setup (Section 7.2)

The last row was not in the original plan. It appeared on the reference build.

Also compare the report with the listing. Section 5 (memory): the expected number of sticks and matching part numbers. Section 6 (storage): one or two NVMe disks. The E16 Gen 2 has two M.2 slots, and if two disks appear, both are wiped in Part 5. Section 7 (platform security): TPM specification version begins with 2.0, and Secure Boot is True.

Reference build. Section 3: no firmware key, volume MAK key, activated, no KMS host, no activator traces. Sections 5 to 7: one 32 GB memory stick from an unknown manufacturer where two 16 GB sticks were listed (single-channel operation), one 1 TB NVMe drive, TPM 2.0 present, Secure Boot False.

Stopping point. Record the gate result and the report file name.

5. Clean Install

1. Insert **stick A**. Power on, press **F12** at the logo, and choose the USB (UEFI) entry.
2. In Windows Setup: language, format, and keyboard, then **Install Windows 11**. Tick the box confirming that everything will be deleted.
3. **Product key page**. If the firmware carries a Pro key, Setup skips this page and installs Pro. If it asks, choose "I don't have a product key" and then **Windows 11 Pro**, which is correct only when the decision gate allows it.
4. Accept the license terms.
5. **Disk selection**. Delete every partition on the internal disk or disks:
 - Select each partition on Disk 0 and choose Delete partition until Disk 0 shows only Unallocated space.
 - If Disk 1 exists, delete its partitions too.
 - Select Disk 0's Unallocated space and choose Next. Setup creates its own EFI, MSR, OS, and Recovery partitions.
6. Setup copies files and restarts several times. When the **Oobe region screen** appears, **remove stick A** and stop. Do not click anything.

Deleting every partition, instead of installing over the top, is the difference between "reinstalled Windows" and "known-clean image". Otherwise the seller's recovery partition and `Recovery\OEM` customizations survive and a later reset can restore them.

Reference build. The installation media downloaded on 30 September 2026 installed Windows 11 version 26H2, newer than the 25H2 build that coverage of the media refresh two weeks earlier had described. Media versions move faster than documentation. Record the version actually installed with `winver`. The unit had no firmware key, so Setup asked. The choices were "I don't have a product key" and Windows 11 Pro, and the retail key was entered after setup.

Building the installer (stick A). On the administrator workstation, download the **Media Creation Tool** from Microsoft's official Windows 11 download page and nowhere else. Run it, accept, choose language and Windows 11, choose **USB flash drive**, and select stick A. This erases stick A. Use Microsoft's unmodified media rather than a tool that strips requirements or forces a local account. APDP is the online Oobe, so the full online experience is the goal.

Preparing stick B. Format as **NTFS** and label it. Copy `Inspect-Laptop.ps1` to the root.

6. Device Association & Provisioning

6.1 Wired Connection

Plug in Ethernet. The laptop sits on the region screen with a wired connection.

6.2 Export Device Information

1. Insert stick B (NTFS).
2. On the region screen, press the **Windows key five times quickly**. The **Autopilot menu** opens.
3. Choose **Export device information**. The DeviceLink CSV saves to stick B.
4. Photograph the menu. Remove stick B.

The device also offers a QR code route, which avoids carrying a CSV on USB. The reference build did not use it.

If the Autopilot menu never opens after several attempts with fast presses on the region screen, go to Section 6.6.

6.3 Pre-Associate in Intune

1. Copy the CSV from stick B to the administrator workstation.
2. Intune admin center > Devices > Enrollment > **Device association** > Devices > **Add**.
3. Upload the CSV and choose Next.
4. Assign the device preparation policy `<ORG>-APDP-UserDriven-v1`. Choose Next, then Add.
5. Confirm the device shows **Association state: Pre-associated**.

Assigning the policy to the device here is deliberate. A device-level assignment takes precedence over a user-level one. The laptop receives the right policy whoever signs in, which is the pattern for client fleets, where the technician who unboxes a laptop is not the person who uses it.

6.4 Complete Association on the Laptop

On the laptop, reopen the Autopilot menu if it closed (Windows key five times) and choose **Next**. The device contacts the Autopilot service, proves its identity with the TPM, and writes the tenant marker into UEFI. Expect **Association complete**. In Intune the state changes to **Associated**.

6.5 Run OOB

Continue. Because the device is associated, wired, and on Pro, language, keyboard, license terms, privacy, and the "personal or work" pages are skipped. The work sign-in page appears, branded if Section 3.9 was done.

1. Sign in as `user@contoso.com`, enter the password, and approve MFA.
2. The device preparation screen shows a percentage while the device Entra joins, enrolls, joins the device group, installs Company Portal and Microsoft 365 Apps, and runs the debloat script. Do not touch it. Photograph at intervals and record start and finish times.

3. When it finishes, the user is signed in. Windows Hello setup begins. Set a PIN.
4. The desktop appears. Record the time.

Reference build. Association complete at 22:09, sign-in and APDP start at 22:14, desktop at 22:23. All of the skipped pages were skipped as designed. Windows Hello was configured with a PIN only and no biometric enrollment, per the owner's standing rule.

If setup fails, the error page shows the custom message and a diagnostics link. Save the diagnostics to stick B, photograph the error, and use the troubleshooting table in Part 10. Do not improvise fixes on the device. Most failures are a tenant-side setting.

6.6 Fallback Without Association

Use this only if Sections 6.2 to 6.4 failed. Skip association and continue OOBЕ normally: region, keyboard, network, **Set up for work or school**, sign in as `user@contoso.com`.

APDP still runs, because the policy is assigned to `SG-APDP-Users`. The differences: no device name template (rename in Section 7.4), OOBЕ pages are not trimmed, and Intune marks the device **Corporate** anyway, because APDP enrollment is corporate by default when no corporate identifiers exist in the tenant. Record why association failed.

7. Verification

Verify from the client first and the portal second. The portal lags reality. Open **Terminal (Admin)** for the commands.

7.1 Join State

```
dsregcmd /status
```

`dsregcmd` is the built-in Device Registration tool. `/status` prints the device's identity relationship with Entra ID. Confirm:

- `AzureAdJoined : YES`, `DomainJoined : NO`, `EnterpriseJoined : NO`. This is a pure Entra join.
- `TenantName` is the expected tenant. Record `DeviceId` and never publish it.
- `TpmProtected : YES`. The device key is sealed in the TPM.
- `AzureAdPrt : YES`. The user holds a Primary Refresh Token, the credential that provides single sign-on to Microsoft 365 apps without retyping a password. Without a PRT, single sign-on fails everywhere.

7.2 Edition & Activation

```
Get-CimInstance SoftwareLicensingProduct -Filter "ApplicationID='55c92734-d682-4d71-983e-d6ec3f16059f' AND PartialProductKey IS NOT NULL" |  
    Select-Object Name, Description, PartialProductKey, LicenseStatus
```

This queries the licensing service for the Windows product only. The GUID is the Windows application ID. Expect the Professional edition and `LicenseStatus 1` (licensed). Settings > System > Activation shows the same information in words.

Check the `Description` channel. A retail or OEM channel is a license for this device. `PartialProductKey` should match the last five characters of the key on record.

Reference build. After the clean install, Settings showed **Windows 11 Business subscription - active**. That is Business Premium subscription activation, an upgrade right layered on a qualifying Windows Pro license. It is not a license on its own, and nothing legitimate sat underneath it. A retail Windows 11 Pro key was bought from the Microsoft Store and entered at Settings > System > Activation > **Change product key**, which prompts for administrator credentials (use the LAPS account). Afterward `Description` contained RETAIL and `LicenseStatus` was 1. Store the key in a password manager and file the receipt.

7.3 Local Administrators

```
net localgroup Administrators  
whoami /groups | findstr /i "S-1-12-1 Administrators"
```

- `net localgroup Administrators` lists the local Administrators group. Expect the disabled built-in Administrator, the LAPS account `<org>-lapsadmin` once LAPS has run, and a raw SID beginning `S-1-12-1-` for the Microsoft Entra Joined Device Local Administrator role. Entra roles appear as SIDs because they are cloud objects the local machine cannot resolve to names. With the Section 3.2 toggle at No, the Global Administrator role is not added, so no SID for it appears. The signed-in user is not listed by name. The standard-user setting worked.
- `whoami /groups` shows the groups in the current token. For a standard user, `BUILTIN\Administrators` does not appear in the list. If it does appear, the user is an administrator, and the Section 3.2 toggle or the APDP account type setting needs correcting.

Capture both outputs. They are the clearest least-privilege evidence in the build.

7.4 Device Name

`hostname` should return `<ORG>-<serial>`. After a fallback build: Intune admin center > Devices > the device > **Rename device** > `<ORG>-<serial>` > restart when prompted.

7.5 BitLocker

```
manage-bde -status C:
(Get-BitLockerVolume -MountPoint C:).KeyProtector
```

- `manage-bde` is the BitLocker command-line tool. Expect **Encryption Method: XTS-AES 256**, **Protection Status: Protection On**, and 100 percent encrypted. Encryption may still be in progress.
- The second command lists key protectors. Expect a **Tpm** protector and a **RecoveryPassword**.
- Portal check: Entra admin center > Devices > the device > **BitLocker keys** shows the escrowed key.

Reference build. The drive was at **XTS-AES 128**. Automatic device encryption won the race against the policy.

Check and fix.

1. Confirm the policy itself arrived. If it did not, fix the policy, not the disk.

```
Get-ItemProperty 'HKLM:\SOFTWARE\Policies\Microsoft\FVE' |
    Select-Object EncryptionMethodWithXts0s, EncryptionMethodWithXtsFdv,
    EncryptionMethodWithXtsRdv, OSEncryptionType
```

This reads the BitLocker policy values Intune wrote. `EncryptionMethodWithXts0s = 7` means XTS-AES 256 was delivered. On the reference build the value was 7 while the disk was 128-bit, which is the proof of the race.

2. **Do not decrypt and wait for Intune to re-encrypt.** On the reference build, after `manage-bde -off C:` the drive sat at "Fully Decrypted, 0.0%" for more than 15 minutes. Neither the "BitLocker MDM policy Refresh" scheduled task nor the "BitLocker Encrypt All Drives" task restarted

encryption. The policy had already been processed and Intune has no equivalent of `gpupdate /force` to replay it.

3. From an elevated terminal (LAPS account), decrypt if not already decrypted, then encrypt at the correct strength:

```
manage-bde -off C: # only if still encrypted; wait for "Fully Decrypted" with manage-  
bde -status C:  
manage-bde -on C: -UsedSpaceOnly -EncryptionMethod xts_aes256 -RecoveryPassword -SkipHardwareTest
```

- `-on C:` turns BitLocker on for the OS drive.
- `-UsedSpaceOnly` encrypts only the space in use. It is fast on a fresh install and matches what the policy's encryption type requests.
- `-EncryptionMethod xts_aes256` sets the cipher explicitly.
- `-RecoveryPassword` adds a 48-digit recovery password protector. Confirm the TPM protector is also present with `manage-bde -protectors -get C:`. Protection On requires a startup protector.
- `-SkipHardwareTest` skips the restart-to-test step. Association already proved the TPM.

4. Escrow the new recovery password to Entra. The old key entry stays in Entra and is now stale.

```
$rp = (Get-BitLockerVolume -MountPoint C:).KeyProtector | Where-Object KeyProtectorType -eq  
'RecoveryPassword'  
BackupToAAD-BitLockerKeyProtector -MountPoint C: -KeyProtectorId $rp.KeyProtectorId
```

The first line finds the recovery-password protector. The second uploads it to the device's Entra object.

5. Verify. `manage-bde -status C:` shows **XTS-AES 256** and **Protection On**. The Entra device shows the new key ID under BitLocker keys. Never screenshot the key itself.

A client template should include a control that stops Windows from encrypting before policy arrives, so this check passes without manual work (Part 9).

7.6 Windows LAPS

```
Get-LocalUser  
Get-WinEvent -LogName 'Microsoft-Windows-LAPS/Operational' -MaxEvents 15 |  
Format-Table TimeCreated, Id, Message -Wrap
```

- `Get-LocalUser` should show `<org>-lapsadmin`, enabled.
- The LAPS event log should show the policy processing and a successful backup of the password to Entra ID. If it has not run yet, `Invoke-LapsPolicyProcessing` forces a cycle.
- **Retrieve the password.** Intune admin center > Devices > the device > **Local admin password**, or Entra admin center > the device > Local administrator password recovery, then Show. Note the last-rotated time.
- **Prove it works.** Sign out, choose Other user, enter `.\<org>-lapsadmin` and the retrieved password. It signs in. Sign out.

- **Prove rotation.** Intune admin center > the device > **Rotate local admin password** (remote action). Sync, then retrieve again. The password and the timestamp have changed.

Never screenshot the password.

7.7 Applications & Debloat Result

- Start menu: Company Portal, Word, Excel, Outlook, and Teams are present. In Word, File > Account shows Microsoft 365 Apps for enterprise and Current Channel.
- Debloat proof:

```
Get-ChildItem C:\ProgramData\<OrgName>\Logs\
Get-Content (Get-ChildItem C:\ProgramData\<OrgName>\Logs\Remove-Bloat_*.log | Select-Object -Last 1).FullName
Get-AppxPackage -AllUsers | Where-Object Name -match
'Bing|Xbox|Solitaire|Clipchamp|Getstarted|FeedbackHub' | Select-Object Name
```

The first two commands show that the script ran and what it did. The third should return nothing.

- Copilot proof: the Copilot app and the Microsoft 365 Copilot app are still present and sign in with the work account.
- If an app or script is missing, read
C:\ProgramData\Microsoft\IntuneManagementExtension\Logs\IntuneManagementExtension.log (applications) and AgentExecutor.log (scripts).

7.8 Portal Checks

On the administrator workstation:

- Intune admin center > Devices > Monitor > **Windows Autopilot device preparation deployments**: status Success, applications and scripts listed with results.
- Intune admin center > the device: **Ownership: Corporate**, correct primary user, managed by Intune.
- Entra admin center > the device: join type **Microsoft Entra joined**, member of SG-APDP-Devices.
- Intune admin center > Devices > Enrollment > Device association: **Associated**.

7.9 Finishing the Machine

1. **Time zone.** Settings > Time & language > Date & time. Hiding the privacy page turned location off, so automatic time zone has nothing to work with. Set it by hand.
2. **Windows Update** until clean, restarting as needed. Lenovo publishes ThinkPad firmware and drivers through Windows Update. Check Advanced options > Optional updates > Driver updates. Then confirm no driver problems:

```
Get-PnpDevice -PresentOnly | Where-Object Status -ne 'OK'
```

This lists present devices whose status is not OK. Empty output means every device has a working driver.

3. **Per-user clutter that Pro will not let policy control.** This is the Enterprise-only gap from Section 2.1. Set it by hand once:
 - Settings > Personalization > Start: "Show recommendations for tips, shortcuts, new apps, and more" Off. "Show account-related notifications" Off.
 - Settings > Personalization > Taskbar: Widgets Off.
 - Settings > System > Notifications > Additional settings: clear "Get tips and suggestions" and "Suggest ways to finish setting up".
4. **Lenovo software.** Skip it unless Windows Update leaves a driver gap. If one appears, Lenovo Commercial Vantage, a Store app deployable from Intune, is the managed-fleet choice over the consumer Vantage.
5. **BIOS update.** Download the update for the machine type from pcsupport.lenovo.com. **Suspend BitLocker first.** Otherwise the firmware change alters the TPM measurements and the next boot asks for the recovery key.

```
Suspend-BitLocker -MountPoint C: -RebootCount 1
```

This pauses BitLocker protection for exactly one restart and then resumes it automatically. Run the update, let it restart, and confirm `manage-bde -status C:` shows **Protection On**. A BIOS reset or a Secure Boot change invalidates association. A BIOS update does not.

Reference build. The BIOS update moved from 1.18 to 1.29. Association survived. `winget upgrade --all` failed in the LAPS account's terminal, because App Installer is a per-user package that is not registered for that profile. Windows Update and Intune are the update path on a managed device, so this is not a fault to chase.

Record the finish time and the total.

8. User Applications Through Company Portal

Nothing in this part runs during OOBE. Every application is **Available** in Company Portal, not Required, and is assigned to the **user** group.

8.1 Choosing the App Type

Work down this list and stop at the first type that fits.

Order	Intune app type	Use when
1	Microsoft Store app (new)	The app is in the Store catalog. Intune keeps it updated
2	Store app (new), Win32-in-Store listings	The same search returns a vendor installer that the Store hosts. Watch install behavior (Section 8.3)
3	Windows MSI line-of-business app	The vendor ships a single per-machine MSI
4	Windows app (Win32), packaged with the Win32 Content Prep Tool	Anything else, with install and uninstall commands and a detection rule
5	Installed web app (browser "Install page as app")	No desktop package exists and the web app is enough. Per-user, not managed

8.2 Create a Store App (New)

Intune admin center > Apps > Windows > Create > **Microsoft Store app (new)** > search > select > set install behavior (Section 8.3) > Assignments: **Available for enrolled devices** > SG-APDP-Users > Create.

8.3 Install Behavior: System or User

- **System** installs once for the machine. Right for per-machine installers.
- **User** installs into the signed-in user's profile. Right for per-user installers.

A per-user installer run as System can land in the SYSTEM profile. Intune reports Installed, but the app never appears in the user's Start menu. If that happens, switch to User and reinstall from Company Portal.

8.4 Chrome as an MSI Line-of-Business App

1. Download the Windows 64-bit MSI from chromeenterprise.google/download.
2. Intune admin center > Apps > Windows > Create > **Line-of-business app** > upload the MSI.
3. Name `Google Chrome`, publisher `Google LLC`. **App install context** is greyed out on Device, because Intune read the MSI and found it per-machine only. Set **Ignore app version** to **Yes**, because Chrome updates itself and Intune would otherwise see version drift and try to reinstall the uploaded version. Leave the command line blank. Intune runs MSIs silently.
4. Assignments: Available > `SG-APDP-Users`.

MSI line-of-business is acceptable here and not everywhere. Mixing MSI line-of-business and Win32 apps as **Required** apps during provisioning can make the installs collide. As an Available app after the desktop, it works. For a client template with Required apps, package Chrome as Win32.

8.5 Expect a Delay

New Available assignments took about one hour to appear in Company Portal and in the web portal (portal.manage.microsoft.com). A device sync does not help, because the app list comes from the Intune service. Re-check after 30 to 60 minutes. Troubleshoot only after about 2 hours: assignment target, group membership (not ownership), and signing out and back in to Company Portal.

8.6 Verify

1. As the standard user: Company Portal > Apps > the app > **Install. No UAC prompt** should appear. The Intune Management Extension installs as SYSTEM on the user's behalf.
2. The app appears in Start.
3. Intune admin center > the app > **Device install status** shows **Installed**. The report lags the device by 5 to 15 minutes.
4. Win32 apps install one at a time. "Download pending" on several apps is a queue, not a fault.

8.7 Known Gap

A standard user can still run per-user installers that write only to the user's own profile (%LOCALAPPDATA%). Two desktop applications on the reference build installed this way with no UAC prompt, bypassing Company Portal. Closing the gap requires App Control for Business (formerly WDAC) or AppLocker. Record the decision to close it or to accept it.

9. Hardening Backlog

None of these belong in the first session. Each is its own piece of work.

#	Item	Why it waits
9.1	Block personal Windows enrollment (Intune > Enrollment > Device platform restriction > Windows > Personally owned: Block)	Safe only after the first device is confirmed Corporate and Associated. Stops anyone from enrolling a personal device through "Allow my organization to manage my device"
9.2	Defender for Business onboarding (Defender portal > Intune connection > Intune EDR policy)	Included in Business Premium. Defender Antivirus is on by default. Onboarding adds EDR visibility and is its own verification exercise
9.3	Compliance policy , and set "Mark devices with no compliance policy assigned as" to Not compliant	Required before Conditional Access
9.4	Conditional Access: require compliant device , break-glass account excluded, report-only first	Possible once a compliant device exists. Requires turning off Security Defaults, which is its own decision
9.5	Windows Update ring and feature update policy	Pin a version until the next feature release is validated
9.6	Separate administrator account	Keep Global Administrator off the daily account
9.7	Widgets and Windows Hello biometrics off by policy	Settings catalog. Makes a no-biometrics rule enforced rather than a personal habit
9.8	Storage Sense by policy	Replaces third-party cleanup tools on a managed device
9.9	App Control for Business decision	Closes the per-user installer gap in Section 8.7, or records the decision to accept it
9.10	Pre-policy encryption control	So that Section 7.5 passes without manual re-encryption on the next device
9.11	Win32 packaging for applications that have no Store or MSI package	Part of an endpoint automation project

Before any Conditional Access policy: confirm the break-glass account can sign in and exclude it from every policy. Flip "Mark devices with no compliance policy assigned as" to **Not compliant**.

Reference build. The reference laptop reported **Compliant** with no compliance policy assigned at all, because the tenant default marks unassigned devices Compliant. The reference build also left the daily account as a Global Administrator, a lab decision recorded as a compromise, not a recommendation. Production guidance is item 9.6.

10. Troubleshooting Reference

Symptom	Most likely cause	Check
Windows key five times does nothing	Not on the region screen, presses too slow, or the build predates KB5120998	Try again on the region page. Otherwise use Section 6.6
Association fails or "device not found"	CSV not uploaded yet, or BIOS reset or Secure Boot change after export	Intune device association list. Re-export and re-upload
"Assign device association" demands Ethernet	Association requires a wired connection in practice	Move to a wired drop or use a USB-C Ethernet adapter
"Personal or work" page still appears	Association not complete, or the edition is not Pro	Association state in Intune. Section 2 of the inspection report
Sign-in works but APDP never starts	User not in <code>SG-APDP-Users</code> , policy not assigned, MDM scope excludes the user, or the device is registered in classic Autopilot	Group membership, policy assignment, Section 3.3, Intune Windows Autopilot devices list
Error 801C03ED	The user is not allowed to join devices	Section 3.2
Error 80180014	An enrollment restriction blocked the device	Intune > Enrollment failures > failure reason
Error 8018000A	The device is already enrolled by another user	Delete the stale device record in Intune and Entra, retry
Device never lands in <code>SG-APDP-Devices</code>	Intune Provisioning Client is not an owner of the group	Section 3.4. Intune > Monitor > Enrollment time grouping failures
Owner search finds no Intune Provisioning Client	The service principal was never created in a new tenant	Section 3.4, <code>New-MgServicePrincipal</code>
App fails during setup	App not assigned Required to the device group, or not in system context	APDP deployment report. <code>IntuneManagementExtension.log</code>
Debloat script did not run	Not assigned to the device group, not selected in the policy, or ran 32-bit	<code>AgentExecutor.log</code> . Section 3.6 settings
BitLocker on but 128-bit	Encryption started before policy arrived	Section 7.5. Decrypting alone does not trigger re-encryption
BitLocker policy "Succeeded" but the disk disagrees	"Succeeded" means the setting reached the registry	Section 7.5
LAPS account missing, or no password in the portal	Entra LAPS switch off, or the device is not on 24H2 or later	Section 3.2. LAPS event log. <code>Invoke-LapsPolicyProcessing</code>
Business subscription "active" with no real license underneath	Subscription activation is a step-up over Pro, not a base license	Section 7.2. Check the <code>Description</code> channel. Buy a retail key
Available apps missing from Company Portal	First-time propagation, about an hour	Section 8.5
App "Installed" in Intune, not in Start	Per-user installer deployed as System	Section 8.3. Switch install behavior to User
<code>winget</code> fails in the administrator terminal	App Installer not registered for the LAPS profile	Not a fault. Update through Intune
Device "Compliant" with no compliance policy	Tenant default marks unassigned devices Compliant	Part 9

11. Verification Checklist

#	Check	Done
1	Business Premium (and Copilot, if used) assigned to the build user	☐
2	Entra device settings saved, including the Global Administrator toggle and Entra LAPS	☐
3	MDM user scope set, MAM scope None	☐
4	Both groups exist, assigned. Intune Provisioning Client owns the device group	☐
5	Company Portal and Microsoft 365 Apps created, Required to the device group	☐
6	Debloat script uploaded, SYSTEM context, 64-bit, assigned to the device group	☐
7	BitLocker policy saved with "Do not enable BitLocker until recovery information is stored" True	☐
8	LAPS policy saved with the four Automatic Account Management settings	☐
9	Device preparation policy created, assigned to the user group	☐
10	BIOS and firmware finished before association. Supervisor password stored first	☐
11	Inspection report read and the decision gate recorded	☐
12	Clean install with every partition deleted	☐
13	Association state Associated. Policy assigned to the device	☐
14	OOBE completed through to the desktop. Windows Hello set	☐
15	<code>dsregcmd</code> : AzureAdJoined YES, DomainJoined NO, TpmProtected YES, AzureAdPrt YES	☐
16	Windows edition Professional, retail or OEM channel, LicenseStatus 1	☐
17	The signed-in user is not a local administrator	☐
18	Device name matches the template	☐
19	BitLocker XTS-AES 256, Protection On, the current key in Entra	☐
20	LAPS account signs in. Password rotated remotely and the value changed	☐
21	Debloat log present, removed-apps query empty, Copilot present	☐
22	Portal: deployment Success, Ownership Corporate, member of the device group	☐
23	Time zone set, Windows Update clean, no driver problems	☐
24	User applications Available in Company Portal and install without a UAC prompt	☐
25	Hardening backlog items scheduled	☐

Sources

- Microsoft Learn: Windows Autopilot device preparation overview, requirements, and what's new (device association on 27 August 2026; OOB BitLocker change on 14 September 2026). learn.microsoft.com/autopilot/device-preparation/overview
- Microsoft Learn: Device association overview and requirements (KB5120998, TPM 2.0, physical device). learn.microsoft.com/autopilot/device-preparation/device-association/overview
- Microsoft Learn: Enrollment time grouping. learn.microsoft.com/intune/device-enrollment/setup-time-grouping
- Microsoft Learn: Identify devices as corporate-owned. learn.microsoft.com/intune/device-enrollment/add-corporate-identifiers
- Microsoft Learn: Manage local administrators on Microsoft Entra joined devices. learn.microsoft.com/entra/identity/devices/assign-local-admin
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- Lenovo PSREF: ThinkPad E16 Gen 2 (AMD) specifications. psref.lenovo.com

Attribution. Written by Evan Ireland in collaboration with Claude (Anthropic). Several claims in the first draft of the procedure were written from documentation before the build and proved wrong on the device: that the Global Administrator role is always a local administrator, that the 14 September 2026 change would prevent 128-bit BitLocker encryption, which installation media version would be current, and that association would work on Wi-Fi. Each is corrected in place in this document. The rule that follows is to assert only what was verified on the device.